

Name: _____



Estimating numerical calculations

Task A: Estimating using significant figures

1) Fill in the gaps to make the approximations correct. You can only use each number once.

250 200 4 4 900 50 5

- a) $4.78 \times 927 \approx \boxed{} \times 900$
b) $3.892 \times 893 \approx \boxed{} \times \boxed{}$
c) $203.38 \div 4.302 \approx \boxed{} \div \boxed{}$
d) $245.29 \div 47.29 \approx 250 \div \boxed{}$

2) To **estimate** this,

$$19.173 \times 338$$

a) Which of these calculations would you use? Explain your choice.

20×350 20×300 19×350 19×300

b) Which of these gives the closest **estimate**?

c) Which gives the **estimate** that is furthest away?

3) **Estimate** the following, making sure to show your working out.

e.g.

a)
$$\frac{5742 \times 34.7}{875.42}$$

b)
$$(61.35 - 84.2)^2$$

c)
$$\frac{15.3 - 7.81}{0.24}$$

d)
$$\frac{9.6 + 16.51}{19.3 + 9.51^2}$$

e)
$$\frac{11.049 - 4.21}{0.82 - 0.301}$$

f)
$$34^2 - \sqrt{139.82}$$

Task B: Estimation in context

- 1) A science textbook uses comparisons with humans to show the sizes of different animals.

- a) Which of these is a good **estimate** for the width of the hand?

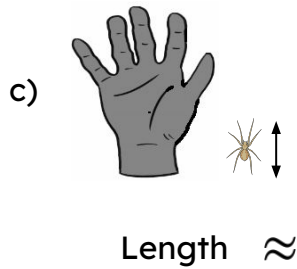
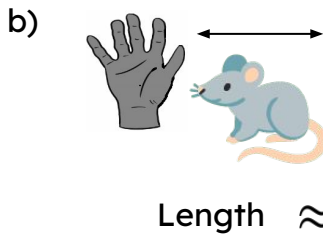
10 cm

25 cm

5 cm



Use your answer to part a to **estimate** the size of the animals below.



- 2) a) What would be a sensible unit of mass to measure each of the items in the table?

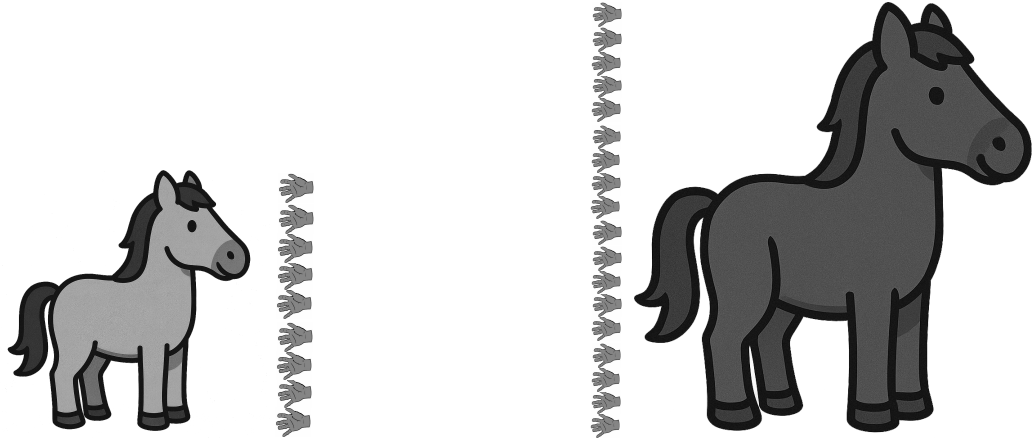
baby	elephant	full suitcase
cargo ship container	newborn puppy	packet of crisps

- b) Each of the values below matches one of the six items. Which is which?

4000	3.2	120	28	19	2797
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- c) An orange weighs approximately 120 g. How many oranges do you think would be the equivalent of one of each of the things in part a?

3) Horses are measured using 'hands' as a unit, as shown below.



a) **Estimate** the height in centimetres of each of the horses.

b) The standard metric equivalent for a hand is 10.2 cm. Does this mean your **estimate** will be too big, too small or just right?

c) How many hands high are you to the top of your head?

d) **Estimate** some other objects using hands as a unit.

